Module 07

Introduction to Relational Model/2

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Module Recap

- Basic notions of modeling introduced
- Attributes and their Types
- Schema and Instance
- Keys and their Categorization
- Languages for Relation Model introduced

Module Objectives

- To understand relational algebra
- To familiarize with the operators of relational algebra

Module Outline

- **Operations**
  - Select
  - Project
  - Union
  - Difference
  - Intersection
  - Cartesian Product
  - Natural Join
- **Aggregate Operations**

Relational Operators

Basic Properties of Relations

A relation is a set. Hence,

- **Ordering of rows / tuples is inconsequential**

A	B
a1	b1
a1	b2
a2	b1
a2	b2

is same as:

A	B
a1	b1
a2	b1
a2	b2
a1	b2

- **All rows / tuples must be distinct**

A	B
a1	b1
a1	b2
a1	b2
a1	b1

is not valid

Select Operation - selection of rows (tuples)

- Relation r
- $\sigma_{A=B \wedge D>5}(r)$

A	B	C	D
α	α	1	7
β	β	23	10

Project Operation - selection of columns (Attributes)

- Relation r
- $\pi_{A,C}(r)$

A	B	C
α	10	1
α	20	1
β	30	1
β	40	2

A	C
α	1
α	1
β	1
β	2

=

A	C
α	1
β	1
β	2

Partha Pratim Das

Union of two relations

- Relation r, s
- $r \cup s$

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

A	B
α	1
α	2
β	1
β	3

Set difference of two relations

- Relation r, s
- $r - s$

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

A	B
α	1
β	1

Set intersection of two relations

- Relation r, s
- $r \cap s$
- **Note:** $r \cap s = r - (r - s)$

A	B
α	1
α	2
β	1

r

A	B
α	2
β	3

s

Joining two relations - Cartesian-product

- Relation r, s
- $r \times s$

A	B
α	1
β	2

r

C	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

s

A	B	C	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

Cartesian-product - naming issue

- Relation r, s
- $r \times s$

A	B
α	1
β	2

r

B	D	E
α	10	a
β	10	a
β	20	b
γ	10	b

s

A	$r.B$	$s.B$	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

Renaming a Table

- Allows us to refer to a relation, (say E) by more than one name.
- $\rho_X(E)$: returns the expression E under the name X
- Relations r
- $r \times \rho_s(r)$

A	B
α	1
β	2

r

$r.A$	$r.B$	$s.A$	$s.B$
α	1	α	1
α	1	β	2
β	2	α	1
β	2	β	2

Composition of Operations

- Can build expressions using multiple operations
- Example: $\sigma_{A=C}(r \times s)$
- $r \times s$
- $\sigma_{A=C}(r \times s)$

A	B	C	D	E
α	1	α	10	a
α	1	β	10	a
α	1	β	20	b
α	1	γ	10	b
β	2	α	10	a
β	2	β	10	a
β	2	β	20	b
β	2	γ	10	b

Joining two relations - Natural Join

- Let r and s be relations on schemas R and S respectively. Then, the **natural join** of relations R and S is a relation on schema $R \cup S$ obtained as follows:
- Consider each pair of tuples t_r from r and t_s from s.
- If t_r and t_s have the same value on each of the attributes in $R \cap S$, add a tuple t to the result, where
  - t has the same value as t_r on r
  - t has the same value as t_s on s

Natural Join Example

- Relations r, s:
- Natural Join

$$r \bowtie s = \pi_{A,r.B,C,r.D,E}(\sigma_{r.B=s.B \wedge r.D=s.D}(r \times s))$$

A	B	C	D
α	1	α	a
β	2	γ	a
γ	4	β	b
α	1	γ	a
δ	2	β	b

r

A	B	C	D	E
α	1	α	a	α
α	1	α	a	γ
α	1	γ	a	α
α	1	γ	a	γ
δ	2	β	b	δ

Aggregation Operators

Aggregate Operators

- Can we compute:
  - **SUM**
  - **AVG**
  - **MAX**
  - **MIN**

Notes about Relational Languages

- Each query input is a table (or set of tables)
- Each query output is a table
- All data in the output table appears in one of the input tables
- Relational Algebra is **not** Turing complete

Summary of Relational Algebra Operators

Symbol (Name)	**Example of Use**	**Description**
σ (Selection)	$\sigma_{\text{salary} \geq 85000}(\text{instructor})$	Return rows of the input relation that satisfy the predicate.
π (Projection)	$\pi_{\text{ID,salary}}(\text{instructor})$	Output specified attributes from all rows ... Remove duplicates.
$\times$ (Cartesian Product)	instructor $\times$ department	Output all possible combinations of rows from the two relations.
$\cup$ (Union)	$\pi_{name}(\text{student}) \cup \pi_{name}(\text{instructor})$	Output the union of tuples from the two input relations.
$-$ (Set Difference)	$\pi_{name}(\text{student}) - \pi_{name}(\text{instructor})$	Output the set difference of tuples.
$\bowtie$ (Natural Join)	instructor $\bowtie$ department	Output pairs of rows that agree on values of common attributes.

Module Summary

- Introduced relational algebra
- Familiarized with the operators of relational algebra

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